

Powering AI in a Climate-Stressed World

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Executive Synthesis

The energy transition has acquired three new constraints in the past six months, and they are not arriving on the same clock. The first is the AI load curve: the International Energy Agency's base case now projects global data centre electricity demand at around 945 TWh by 2030, more than doubling the 2024 baseline of 415 TWh, with AI as the dominant driver ([IEA, Apr 2025](#)). The second is the transmission deficit: roughly 10,300 projects representing 1,400 GW of generation and 890 GW of storage were sitting in US interconnection queues at end-2024, with the typical project taking 55 months from request to commercial operation ([Lawrence Berkeley National Laboratory, Dec 2025](#)). The third is the insurability frontier: 2025 was the sixth consecutive year insured natural catastrophe losses exceeded \$100 billion, with Swiss Re putting the total at \$107 billion and the Los Angeles wildfires alone contributing \$40 billion, the largest insured wildfire event on sigma records ([Swiss Re Institute, Mar 2026](#)).

This cycle's evidence supports three structural shifts. First, the cost of AI compute is being internalised into the cost of capital across capacity markets: PJM's 2027/2028 auction cleared at the \$333.44/MW-day price cap, the third record auction in succession, with data centres responsible for around 40% of total auction costs by the market monitor's reckoning ([Utility Dive, Dec 2025](#); [Utility Dive, Dec 2025](#)). Second, the regulatory toolkit for moving electrons has finally arrived (FERC Order 1920 in May 2024, the European Commission's Affordable Energy Action Plan in February 2025, ENTSO-E's TYNDP 2026 draft portfolio of 178 transmission and 49 storage projects) but every clock on the toolkit runs in years, while the load arrives in months ([FERC, May 2024](#); [European Commission, Feb 2025](#); [ENTSO-E, Oct 2025](#)). Third, climate physical risk is moving from the prudential periphery to the credit and insurance core: California's

FAIR Plan grew to nearly \$700 billion of insured property by September 2025, a 317% increase since 2021, and First Street estimates climate risks could erase \$1.4 trillion of US property value over 30 years ([California Department of Insurance](#), Feb 2026; [First Street](#), Feb 2025).

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For the reader, the consequence is that energy, infrastructure and climate cannot be planned as separate workstreams in the next twelve months. The Talen-AWS Susquehanna deal (1.92 GW, \$18 billion, through 2042) and Constellation's Microsoft Three Mile Island restart (835 MW, around \$1.6 billion, online by 2028) are not three isolated nuclear stories: they are evidence that hyperscalers are bypassing both the queue and the climate-risk re-pricing by buying behind-the-meter firm clean power outright ([POWER Magazine](#), Jun 2025; [Constellation](#), Sep 2024). This report adopts a posture of *integrated capital planning*: every capital decision must now be tested against compute demand, grid interconnection timing, climate physical-loss repricing, and firm-clean supply availability in a single matrix, not four. The reader who does this will pay less to connect, less to insure, and less to source clean firm power than the reader who does not.

Where this analysis could be wrong

The central assumption is that the four constraints (AI compute, transmission, climate physical loss, firm-clean supply) tighten together over the next 18 months rather than resolving independently. If hyperscaler capex moderates materially before end-2026, if FERC Order 1920 and PJM's behind-the-meter colocation reforms accelerate queue clearance rates by 30% or more inside two years, and if the 2026 Atlantic hurricane season and Western US fire season deliver below-average insured losses, the integrated planning posture is overstated and the four can be resolved sequentially. The single observable that would force a material revision is hyperscaler capex guidance through Q3 2026 earnings: a sustained 15%+ guidance reduction across Microsoft, Amazon, Google and Meta would be the disconfirming signal that the demand curve is breaking.